Classical Field Theory — Exercise Sheet 1

Local actions

Hints and answers

Use Lecture 1's sign $E_a = \partial L/\partial\phi^a - \partial_\mu p_a^\mu$. All fields and prescribed sources in this sheet are smooth. The first three problems use only differentiation, integration by parts and the fundamental lemma. The final problem also uses finite-dimensional linear ODEs.

Exercise 1.1 (A complex source and two real variations). On $\mathbb{R}^{1,1}$ let $\Phi = \phi_1 + \mathrm{i}\phi_2$ be a complex field, $J = J_1 + \mathrm{i}J_2$ a prescribed complex source, and $m \geq 0$. The density is

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\Phi_x|^2 - m^2|\Phi|^2\big) + \mathrm{Re}(\bar{J}\Phi).$$

For the action, take any compact region $D \subset \mathbb{R}^{1,1}$ and variations H whose support lies in the interior of D.

(a) Compute both real momentum currents and Euler expressions. Express the stationarity condition as one complex equation and derive the first variation for a compactly supported complex variation H.

(b) Find the source for which $\Phi(t,x) = (t^2+1) + \mathrm{i}(t^3 - x^2)$ is stationary.

(c) If only real-valued variations H are tested, must the complex equation hold? Give a counterexample with $m = 0$ and $J = 0$.

Hint. Expand the real part of $\bar{J}\Phi$ before differentiating. For compact variations, use Lecture 1 separately for ϕ_1 and ϕ_2. Real and imaginary tests detect different Euler expressions.

Answer. The equation is $(\Box + m^2)\Phi = J$. The source in the second part is $2 + m^2(t^2+1) + \mathrm{i}\big(6t + 2 + m^2(t^3 - x^2)\big)$. Real tests alone fail, for example at $\Phi = \mathrm{i}t^2$.

Exercise 1.2 (Natural conditions at the two spatial ends). Let $T, \ell > 0$, $m \geq 0$ and $\beta_0, \beta_\ell \geq 0$. On the rectangle $D = [0,T] \times [0,\ell]$, a real field has action

$$S_D[\phi] = \int_0^T \int_0^\ell \left(\frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2 + J\phi\right) \mathrm{d}x\mathrm{d}t \\ - \frac{1}{2}\int_0^T \big(\beta_0\phi(t,0)^2 + \beta_\ell\phi(t,\ell)^2\big)\mathrm{d}t,$$

where J is prescribed. Variations vanish near $t = 0, T$ and are unrestricted at $x = 0, \ell$.

(a) Integrate coordinate by coordinate and prove that stationarity is equivalent to the bulk equation and two Robin boundary conditions. Include the case $\beta_0 = \beta_\ell = 0$.

(b) For $\ell = 1$, $m = 0$, $J = 2$ and $\beta_0 = \beta_\ell = 1$, find the unique time-independent stationary field.

Hint. The spatial integration gives $[-\phi_x H]_{x=0}^{x=\ell}$. First use interior tests, then choose independent traces at the two ends. The static sourced equation is an elementary second-order ODE.

Answer. The conditions are $\phi_{tt} - \phi_{xx} + m^2\phi = J$, $\phi_x(t,0) = \beta_0\phi(t,0)$ and $\phi_x(t,\ell) = -\beta_\ell\phi(t,\ell)$. The static field in the second part is $1 + x - x^2$.

Exercise 1.3 (Lorentz changes of coordinates). The Lorentz group in two coordinates is the matrix group

$$O(1,1) = \{\Lambda \in GL(2,\mathbb{R}) : \Lambda^{\mathsf{T}} \eta \Lambda = \eta\}, \qquad \eta = \operatorname{diag}(1,-1).$$

For $|v| < 1$, set

$$\gamma_v = (1-v^2)^{-1/2}, \qquad \begin{pmatrix} t' \\ x' \end{pmatrix} = \Lambda_v \begin{pmatrix} t \\ x \end{pmatrix}, \qquad \Lambda_v = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}.$$

(a) Verify that $\Lambda_v \in O(1,1)$ and $\det \Lambda_v = 1$.

(b) For a scalar field, set $\phi'(t',x') = \phi(t,x)$ at corresponding points. Use the chain rule to prove invariance of $\partial_t^2 - \partial_x^2$ and of $\frac{1}{2}(\phi_t^2 - \phi_x^2) - V(\phi)$, where V is any smooth function. Conclude equality of the scalar actions on corresponding compact regions.

Hint. The inverse boost has velocity $-v$. Express both primed derivatives in unprimed derivatives before subtracting the squares. The action also uses the coordinate Jacobian.

Answer. The primed derivatives are $\partial_{t'} = \gamma_v(\partial_t + v\partial_x)$ and $\partial_{x'} = \gamma_v(v\partial_t + \partial_x)$. Their squared difference and the scalar kinetic difference are unchanged; the Jacobian is one.

Exercise 1.4 ⋆. Let $N \geq 3$, $h_{\mathrm{lat}} = 2\pi/N$, and let $q_j(t)$, $j = 0, \dots, N-1$, be real coordinates with periodic indexing. The chain Lagrangian is

$$L_h = h_{\mathrm{lat}} \sum_{j=0}^{N-1} \left[\frac{1}{2}\dot{q}_j^2 - \frac{1}{2h_{\mathrm{lat}}^2}(q_{j+1} - q_j)^2 - \frac{1}{2}m^2 q_j^2 \right], \qquad m \geq 0.$$

(a) Derive its equations by varying every q_j.

(b) Let $T > 0$ and let ϕ be a given smooth periodic solution of $\phi_{tt} - \phi_{xx} + m^2\phi = 0$ on $[0,T] \times \mathbb{T}$. Assume $M_4 \geq 0$ and $|\partial_x^4 \phi| \leq M_4$ there. Prove that sampling this field in the chain equation produces a residual bounded by $h_{\mathrm{lat}}^2 M_4/12$ at every node.

(c) Give the chain the exact sampled initial position and velocity. Set $e_j = q_j - \phi(t, jh_{\mathrm{lat}})$, $D_h e_j = (e_{j+1} - e_j)/h_{\mathrm{lat}}$ and $\|w\|_h^2 = h_{\mathrm{lat}} \sum_j w_j^2$. With $A_h = \sqrt{2\pi} M_4 h_{\mathrm{lat}}^2/12$, prove, for $0 \leq t \leq T$,

$$\left(\|\dot{e}(t)\|_h^2 + \|D_h e(t)\|_h^2 + m^2 \|e(t)\|_h^2 \right)^{1/2} \leq A_h t, \qquad \|e(t)\|_h \leq \frac{1}{2} A_h t^2.$$

The finite linear chain has a global solution by the linear ODE theorem. No existence theorem for the continuum equation is assumed beyond the given ϕ.

Hint. Each spring contributes to two coordinate variations. For consistency, add Taylor formulas at $x + h_{\mathrm{lat}}$ and $x - h_{\mathrm{lat}}$, with fourth-derivative remainder. For convergence, differentiate the discrete quadratic error energy and use periodic summation by parts. Regularise its square root at zero.

Answer. The equations are $\ddot{q}_j - (q_{j+1} - 2q_j + q_{j-1})/h_{\mathrm{lat}}^2 + m^2 q_j = 0$. The stated estimates give order h_{lat}^2 convergence in the sampled norms on every fixed time interval, including $m = 0$.